Computing the MD5 Hash

1. **Python**

Syntax:

import hashlib

md5Hash = hashlib.md5("*password_to_hash*".encode()).hexdigest()

Example: To calculate the MD5 Hash of the password **abc12**

md5Hash = hashlib.md5("abc12".encode()).hexdigest()

The value of **md5Has**h is **b2157e7b2ae716a747597717f1efb7a0** .

Explanation: You cannot directly apply the **md5** hash to a string. The methods/functions in **hashlib** operate on byte data, so the string must be converted to a byte format prior to hashing. Translating the string into a sequence of bytes is called **encoding**, and **str** objects have an encode method. Additionally, the hashing algorithms return a hash object. To display the hash as a hex value, use the **hexdigest()** method (the output of this method is a string of hex characters).

2. **Linux (command line)**

Syntax:

echo -n "password_to_hash" | md5sum

Example: To calculate the MD5 Hash of the password **abc12 echo**

-n "abc12" | md5sum

The value of **md5Has**h is **b2157e7b2ae716a747597717f1efb7a0** .

Explanation: The **md5sum** command is primarily designed to compute the MD5 hash of file contents (e.g., **$ md5sum file.txt**). Additional steps are needed to accept the string from the keyboard.

- The **echo** command prints the text that follows it to the terminal. Here it's printing the string "**abc12"**.
- The **-n** option for echo prevents the addition of a newline character at the end of the output.
- The pipe operator **(|)** takes the output from the command on the left (**echo -n "password_to_hash"**) and uses it as the input for the command on the right (**md5sum**).

When you run the command, the **echo -n "password_to_hash"** part outputs the string without a newline. This output is then piped into the **md5sum** command, which computes the MD5 hash of the provided string.

3. Windows (command line)

The certutil command used here only works on a file. The string (password) to hash should be placed in a text file. Be sure NOT to have a newline character at the end of the string (don't hit return).

Syntax:

certutil -hashfile *filename_to_hash.txt* MD5

Example: To calculate the MD5 Hash of the password **abc12** that is saved in the file Password.txt:

certutil -hashfile Password.txt MD5

The hash result is **b2157e7b2ae716a747597717f1efb7a0** .

Explanation: **certutil** is a built-in Windows command-line tool primarily used for managing certificates and cryptographic functions. However, it also has a handy feature for computing file hashes using the **-hashfile** option. You can replace MD5 with other algorithms like SHA1, SHA256, or SHA512:

4. macOS (command line)

Syntax:

echo -n "hello" | md5

Example: To calculate the MD5 Hash of the password **abc12**t: **echo**

-n "hello" | md5

The hash result is **b2157e7b2ae716a747597717f1efb7a0** .

Explanation: This command is very similar to the Linux command.

- The **echo** command prints the text that follows it to the terminal. Here it's printing the string "**abc12"**.
- The **-n** option for echo prevents the addition of a newline character at the end of the output.
- The pipe operator **(|)** takes the output from the command on the left (**echo -n "password_to_hash"**) and uses it as the input for the command on the right (**md5**).

When you run the command, the **echo -n "password_to_hash"** part outputs the string without a newline. This output is then piped into the **md5** command, which computes the MD5 hash of the provided string.

5. Online md5 hash calculator

When hashing a password, be sure the string does NOT include the newline character. You will NOT get the correct hash value.

For your online **md5** calculator, the **md5** hash of **abc12** should produce

b2157e7b2ae716a747597717f1efb7a0

6. Chatgpt

Ask chatgpt the basic question : “What is the md5 hash of “password”? ”
If you are trying to find hash of “abc12”, you should be typing “What is the md5 hash of abc12?
In Chatgpt, the md5 hash of abc12 should produce
b2157e7b2ae716a747597717f1efb7a0

Chatgpt uses python in the background to provide you with this result